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Predicting Term Deposit Subscriptions with Machine Learning

ML Final Project Document

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1 Problem description

1.1 Context and Motivation

Direct marketing campaigns constitute a significant cost center for banking institutions. The ability to anticipate whether a client will subscribe to a financial product, such as a term deposit, before making contact reduces operational costs and improves customer experience by avoiding unsolicited contact. This problem is naturally defined as a **binary classification task**, where the goal is to predict client subscription from demographic, financial, and campaign-related information.

1.2 Dataset

The dataset used in this project is the **Bank Marketing Dataset**, originally collected by a Portuguese banking institution [1] and made publicly available through Kaggle [2]. It contains records of direct phone-call marketing campaigns conducted between May 2008 and November 2010. We note that this Kaggle distribution differs from the canonical UCI **bank-full.csv** of [1]: both share the same 45,211 records, but the version used here is less imbalanced, with a positive rate of 17.5% against the 11.7% of the original. All results reported in this work refer to this version.

The dataset consists of **45,211 instances** and **16 input features**, which can be grouped into four categories:

- **Client demographics:**

- **age** — client’s age (numeric).
- **job** — type of job (categorical).
- **marital** — marital status (categorical).
- **education** — education level (categorical).

- **Financial status:**

- **default** — whether the client has credit in default (binary).
- **balance** — average yearly account balance in euros (numeric).
- **housing** — whether the client has a housing loan (binary).
- **loan** — whether the client has a personal loan (binary).

- **Campaign information:**

- **contact** — contact communication type (categorical).
- **day, month** — day and month of the last contact.
- **duration** — duration of the last contact in seconds (numeric).
- **campaign** — number of contacts made during this campaign (numeric).

- **Previous-campaign information:**

- **pdays** — days since the client was last contacted in a previous campaign (numeric; -1 if not previously contacted).
- **previous** — number of contacts before this campaign (numeric).
- **poutcome** — outcome of the previous campaign (categorical).

The target variable y is binary: **yes** if the client subscribed to a term deposit, **no** otherwise.

For predictive modelling, the feature set is transformed in two ways (detailed in Section 2.5 and the preprocessing subsection): the leaky **duration** feature is removed, and a binary **prev_contacted** indicator is engineered from the **pdays** sentinel value (-1 , meaning “never previously contacted”). The resulting predictive set therefore also comprises 16 variables, which expand to 51 columns after one-hot encoding of the categorical features.

1.3 Challenges

This dataset presents several non-trivial challenges that make it a realistic evaluation setting:

- **Class imbalance:** the 4.7:1 ratio (Section 1.2) makes standard accuracy a misleading metric — a trivial all-no classifier already scores 82.5% — so class weighting and decision-threshold tuning are required.
- **Mixed feature types:** the dataset combines numerical and categorical variables, requiring appropriate preprocessing before model training.
- **Structured “unknown” values:** several categorical features contain **unknown** entries at very different rates, from **job** (1.1%) and **education** (7.3%) to **contact** (24.0%) and **poutcome** (76.2%). The high rate in **poutcome** is not random missingness but structurally identifies clients with no prior campaign history, so these values must be handled deliberately rather than imputed.
- **Sentinel encoding:** **pdays** uses the value -1 to denote clients never contacted before (73.7% of records), a non-trivial encoding that must be separated from genuine numerical distances.
- **Potential data leakage:** the feature **duration** (last call duration) is highly predictive ($r = 0.372$) but is only known after the call ends, making it unsuitable for realistic pre-call prediction. Its inclusion must be explicitly discussed and, in this work, it is excluded from all predictive models.

2 Methodology

This section describes the complete machine learning pipeline applied to the Bank Marketing dataset, covering data preprocessing, unsupervised structural analysis, model training and tuning, and evaluation strategy.

2.1 Data Preprocessing

Raw data requires several preprocessing steps before being used for model training.

Encoding. Categorical variables (`job`, `marital`, `education`, `default`, `housing`, `loan`, `contact`, `month`, `poutcome`) were encoded using One-Hot Encoding (OHE). OHE was preferred over ordinal encoding because none of these variables have a natural order, and over target encoding because it would require fitting category-to-target statistics that, if not carefully recomputed inside every cross-validation fold, introduce target leakage. The `handle_unknown='ignore'` option was set to produce zero vectors for categories unseen at inference time, a safe and neutral default.

Scaling. Numerical features were standardised to zero mean and unit variance using `StandardScaler`. Standardisation is a hard requirement for models whose optimisation is sensitive to feature magnitude: in Logistic Regression and SVM, the regularisation penalty $\|\mathbf{w}\|^2$ treats all dimensions equally only if they are on the same scale; in PCA, variance is dominated by features with larger ranges if data is not standardised. `StandardScaler` was preferred over `MinMaxScaler` because numerical features such as `balance` and `pdays` contain outliers that would compress the majority of values into a narrow range under min-max normalisation.

Missing values. No explicit NaN values were found in the dataset. However, several categorical features contain `unknown` entries. Because `unknown` is an explicit string value rather than a missing entry, it is preserved through OHE as its own binary indicator; the most-frequent imputer included in the categorical pipeline therefore acts only as a safety net for genuinely absent values (of which there are none) and never overwrites `unknown`. Treating it as a separate category, rather than imputing it, is supported by its prevalence: while `job` (1.1%) could reasonably be imputed, `education` (7.3%), `contact` (24.0%), and especially `poutcome` (76.2%) have proportions too large to treat as random missingness. In particular, `poutcome = unknown` systematically identifies clients with no prior campaign history, a structurally distinct and potentially informative group whose signal would be destroyed by imputation.

Class imbalance. Given the 4.7:1 imbalance ratio, and to address it without introducing synthetic samples, all classifiers that support it were trained using the to its frequency:

$$w_c = \frac{N}{K \cdot N_c} \quad (1)$$

where N is the total number of samples, K is the number of classes, and N_c is the number of samples in class c . The boosting models (`GradientBoosting`, `AdaBoost`) do not expose a class-weight parameter; for these, imbalance is handled at evaluation time through

threshold-independent metrics (AUPR, ROC-AUC) and explicit decision-threshold tuning (Section 2.5).

Pipeline. All preprocessing steps were encapsulated in a single `scikit-learn` `ColumnTransformer`, which is cloned and *re-fitted inside every cross-validation fold*, so that no scaling or encoding statistic is ever estimated on data used for validation. This eliminates the subtle preprocessing leakage that arises when the transformer is fitted once on the full training set.

Data split. The dataset was divided into 80% training and 20% test sets. Given the dataset size of 45211 instances, a 20% test partition yields approximately 9000 held-out samples, sufficient to obtain stable metric estimates. A larger test set was not necessary since model selection relies on cross-validation rather than the test set, which is reserved exclusively for final evaluation. Stratified sampling was applied to preserve the positive rate across both splits.

2.2 Feature Analysis and Selection

Prior to model training, the association between each input feature and the target variable was quantified. For numerical features, the point-biserial correlation coefficient was computed. For categorical features, Cramér’s V statistic was used, based on the χ^2 test of independence:

$$V = \sqrt{\frac{\chi^2}{N \cdot (k - 1)}} \quad (2)$$

where N is the sample size and k is the smaller of the number of rows or columns in the contingency table.

Feature exclusion. All features were first assessed for temporal validity, that is, whether they are observable before a call is made. The feature `duration`, despite presenting by far the highest correlation with the target ($r = 0.372$, $p \approx 0$), represents the length of the last phone call and is therefore only known *after* the call has taken place. Including it constitutes data leakage, as the model would exploit information unavailable at prediction time; the magnitude of this effect is quantified in Section 4.3. It was therefore excluded from all predictive experiments. The remaining features were confirmed to be available prior to client contact and are treated as legitimate predictors.

Association analysis. Among categorical variables, `month` ($V = 0.305$), `job` ($V = 0.201$), `poutcome` ($V = 0.181$) and `housing` ($V = 0.178$) show the strongest legitimate associations; among numerical features, `day` ($r = -0.115$) and `campaign` ($r = -0.098$) are the most associated. No filter-based feature selection was applied; instead, feature relevance was analysed post-hoc through ensemble feature importances (impurity-based and permutation), providing a data-driven ranking without discarding potentially useful variables a priori.

2.3 Unsupervised Structural Analysis

Principal Component Analysis (PCA). PCA was applied to the standardised training data to study its variance structure. Because one-hot encoding spreads variance across many sparse binary columns, dimensionality reduction is modest: 20 components are required to retain 90% of the variance and 27 to retain 95% (out of 51 features). The 90%-variance projection (20 components) was used as input for clustering, and a 2-D projection (PC1 + PC2) was produced to visually inspect class separability.

Clustering. Two clustering algorithms were applied on the PCA-reduced data:

- **K-Means:** the number of clusters k was selected by combining the Elbow Method (inertia) and the Silhouette Score across $k \in \{2, \dots, 10\}$.
- **Gaussian Mixture Models (GMM):** the number of components was selected by minimising the Bayesian Information Criterion (BIC).

In both cases, cluster assignments were compared against the true labels using the Adjusted Rand Index (ARI) to assess whether the unsupervised structure aligns with the supervised signal.

2.4 Model Training and Tuning

Six model families were trained, covering the main paradigms studied in the course. All models that support it used `class_weight='balanced'`.

Baseline. A stratified dummy classifier provided a lower-bound reference to contextualise all other models.

Logistic Regression. An ℓ_2 -regularised logistic regression was trained. The regularisation strength was swept on a logarithmic scale, $C \in \{10^{-4}, \dots, 10^3\}$ (15 points), to study its effect on the coefficient norm and on generalisation. An ℓ_1 variant and a Lasso path on the binary target were additionally used to illustrate coefficient shrinkage and sparsity.

Support Vector Machine (SVM). Linear, polynomial and RBF kernels were first compared by cross-validation. A kernel SVM scales roughly as $\mathcal{O}(n^2)$, and obtaining calibrated probabilities (`probability=True`, required for a uniform `predict_proba` interface) adds an internal cross-validation that makes a full-data fit impractical; SVM was therefore trained on a stratified subsample of 8,000 instances. The RBF kernel was then tuned by Grid Search over $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$ with 3-fold stratified cross-validation, optimising AUPR.

Random Forest. A bagging ensemble of 300 decision trees was trained on the full training set. Impurity-based and permutation feature importances were extracted for interpretation.

Gradient Boosting. A sequential boosting ensemble of 200 estimators (`GradientBoostingClassifier`) was trained, and its feature importances compared against Random Forest.

AdaBoost. An AdaBoost classifier with 200 estimators was trained as an additional boosting model.

2.5 Evaluation Strategy

Metrics. Given the class imbalance, accuracy is not used as a selection criterion. The primary ranking metric is **AUPR**, complemented by ROC-AUC, the F1-score of the positive (minority) class, and balanced accuracy; the per-class precision/recall breakdown (including the macro average) is reported via the classification report. All final metrics are computed on the held-out test set.

Hyperparameter tuning and nested cross-validation. Hyperparameters were selected by **GridSearchCV** on the training set only, scored by AUPR. To obtain an *unbiased* estimate of generalisation, model comparison was performed with **nested cross-validation**: an outer stratified 5-fold loop measures performance while an inner 3-fold **GridSearchCV** re-tunes the hyperparameters *within every outer fold*, so that no information from model selection leaks into the reported scores. The scalable models (Logistic Regression, Random Forest, Gradient Boosting, AdaBoost) were compared this way; the kernel SVM was excluded from the nested loop, since wrapping its super-quadratic training in the double loop is prohibitively expensive, and was instead studied separately on the 8000-sample subsample (Section 3.4). The test set was never used during selection or tuning.

Decision-threshold tuning. Because the default 0.5 threshold is rarely F1-optimal under imbalance, the operating threshold of the selected model was tuned on *out-of-fold* training probabilities obtained with **cross_val_predict** (no leakage) by maximising F1, and then applied unchanged to the test set.

Statistical testing. The two best models by nested-CV AUPR were compared on their per-fold scores using a paired *t*-test and the non-parametric Wilcoxon signed-rank test ($\alpha = 0.05$). With only five (non-independent) folds these tests are treated as indicative rather than conclusive.

Reproducibility. All experiments used a fixed random seed (**SEED** = 42).

3 Experiments

This section reports the experiments carried out at each phase of the pipeline. All experiments use a fixed seed (`SEED = 42`), `scikit-learn` pipelines in which every preprocessing step is re-fitted inside each cross-validation fold, and AUPR as the primary selection metric, given the class imbalance.

3.1 Baselines

Two trivial baselines contextualise all subsequent results. A *most-frequent* classifier reaches 82.5% accuracy but an F1 of 0.000 on the positive class, confirming that accuracy is vacuous here. A *stratified* dummy classifier obtains 0.714 accuracy and 0.171 F1. Any useful model must clearly exceed the latter in F1 and AUPR.

3.2 Phase I — Unsupervised Structural Analysis

PCA. Applied to the 51 one-hot-encoded, standardised features, PCA shows that variance is highly distributed: 20 components are needed to retain 90% of the variance and 27 to retain 95%. This is expected, since one-hot encoding spreads variance across many sparse binary columns; PCA is therefore used here for visualisation and to denoise the clustering space rather than for aggressive dimensionality reduction. The 2-D projection onto the first two principal components shows no linear separation of subscribers from non-subscribers, an early indication that the problem is genuinely hard.

Clustering. K-Means was run for $k \in \{2, \dots, 10\}$; the silhouette score is maximised at $k = 2$ (silhouette = 0.282), a modest value indicating weak but non-negligible cluster structure. A Gaussian Mixture Model selected by BIC/AIC kept favouring the maximum number of components, further confirming the absence of a small number of well-separated groups.

Clusters vs. target. Comparing the $k = 2$ K-Means assignment against the true labels yields an Adjusted Rand Index of only 0.064. The two clusters do differ in subscription rate (25.4% in the smaller cluster of 8979 clients vs. 15.0% in the larger cluster of 27189), so the unsupervised structure carries a faint signal, but it is driven mainly by demographic and contact patterns rather than by the rare subscription outcome. We conclude that clustering is informative as exploratory analysis but cannot substitute for a supervised classifier.

3.3 Phase II — Linear and Regularised Models

Effect of regularisation. An ℓ_2 -regularised logistic regression was evaluated along a logarithmic grid $C \in \{10^{-4}, \dots, 10^3\}$. As C increases (weaker regularisation), the coefficient norm $\|\mathbf{w}\|$ grows monotonically, while the cross-validated F1 peaks at $C = 0.1$ (CV-F1 = 0.462) and then plateaus, illustrating the classic bias-variance trade-off: too much regularisation under-fits, too little adds variance without improving generalisation.

ℓ_1 vs. ℓ_2 and sparsity. Comparing the two penalties at the selected strength, the Ridge-like (ℓ_2) model attains CV ROC-AUC 0.7755 keeping all 51 coefficients non-zero, whereas the Lasso-like (ℓ_1) model attains an essentially identical CV ROC-AUC 0.7754 while zeroing 5 features (46/51 non-zero). A full Lasso path confirms progressive feature elimination, down to zero surviving features at the sparsest end. Since ℓ_1 matches ℓ_2 performance with a sparser, more interpretable model, it is the preferable linear configuration.

Coefficient interpretation. This regularisation study selects C by CV-F1; for the final model comparison (Section 4) the logistic regression is re-tuned by AUPR, the two configurations differing only in C and yielding practically identical test performance (AUPR 0.437 vs. 0.438). On the held-out test set the model reaches ROC-AUC 0.779 and AUPR 0.437. Its largest positive coefficients correspond to `poutcome = success`, contact by `cellular`, and certain months, while the largest negative coefficients align with `unknown/telephone` contact and months with historically low conversion. These drivers agree with the EDA associations (Cramér’s V), lending face validity to the model.

3.4 Phase III — Kernel Methods (SVM)

Because a kernel SVM scales roughly as $\mathcal{O}(n^2)$ and the calibrated-probability fit (`probability=True`) adds an internal cross-validation that makes full-data training impractical, models were trained on a stratified subsample of 8000 instances. The three kernels perform almost identically in 5-fold cross-validation (Table 1), with no kernel offering a decisive advantage. The RBF kernel was then tuned by grid search over $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$ (3-fold CV, AUPR), selecting $C = 1$, $\gamma = 0.01$ (CV-AUPR 0.432; the small gap with Table 1 reflects the 3-fold grid search versus the 5-fold kernel comparison). On the test set the tuned RBF-SVM reaches ROC-AUC 0.778 and AUPR 0.435, on par with logistic regression but at substantially higher computational cost.

Kernel	CV-AUPR	CV ROC-AUC	CV F1
Linear	0.435	0.771	0.456
Polynomial	0.434	0.763	0.463
RBF	0.435	0.775	0.469

Table 1: SVM kernel comparison on the 8000-sample subset (5-fold CV).

3.5 Phase IV — Ensemble Methods

Three ensembles were trained on the full training set with their default configuration (Random Forest: 300 trees; Gradient Boosting and AdaBoost: 200 estimators) and compared on performance, training time, and feature importance (Table 2). Gradient Boosting achieves the best AUPR (0.460) and ROC-AUC (0.789); AdaBoost is very close (0.458 / 0.786); Random Forest trails on AUPR (0.423) but is by far the fastest to train (1.9s). Boosting therefore dominates bagging on this problem. These figures use default hyperparameters; the tuned configurations, reported in Section 4, narrow the gap (notably for Random Forest).

Model	Train time (s)	ROC-AUC	AUPR
Random Forest	1.9	0.761	0.423
Gradient Boosting	12.9	0.789	0.460
AdaBoost	6.9	0.786	0.458

Table 2: Ensemble comparison (default configuration, held-out test set).

Feature importance. Impurity-based and permutation importances broadly agree. The most influential features are `age`, `campaign`, `day`, `contact = unknown`, and the months `aug` and `nov`. Interestingly, the multivariate importance ranking elevates continuous features (`age`, `campaign`, `day`) above some variables that scored highest in univariate association (`month`, `poutcome`), reflecting that the ensembles exploit interactions and conditional effects rather than marginal correlation alone.

3.6 Phase V — Model Selection via Nested Cross-Validation

Unbiased comparison. The four scalable models were compared with nested cross-validation (outer 5-fold, inner 3-fold `GridSearchCV`, AUPR-scored); the kernel SVM was excluded due to its cost and evaluated separately (Section 3.4). Results are reported as mean $\pm$ std over the outer folds (Table 3). Gradient Boosting obtains the highest nested-CV AUPR (0.466 ± 0.013), narrowly ahead of AdaBoost (0.464 ± 0.014), with Logistic Regression and Random Forest behind. Crucially, the nested-CV estimate for Gradient Boosting (0.466) closely matches its later held-out test AUPR (0.460), confirming that the selection procedure is not optimistically biased. Training times also reveal the cost of the unbiased estimate: the boosting models require roughly 26–39 s and Random Forest 21 s, versus only 3 s for logistic regression.

Model	Nested-CV AUPR	Time (s)
Gradient Boosting	0.466 ± 0.013	26
AdaBoost	0.464 ± 0.014	39
Logistic Regression	0.450 ± 0.015	3
Random Forest	0.438 ± 0.011	21

Table 3: Nested cross-validation comparison (outer 5-fold AUPR).

Statistical significance. The two best models (Gradient Boosting vs. AdaBoost) were compared on their per-fold AUPR with a paired t -test ($t = 1.02$, $p = 0.367$) and a Wilcoxon signed-rank test ($W = 3.0$, $p = 0.313$). Neither is significant at $\alpha = 0.05$: the two boosting models are statistically indistinguishable on this dataset, and Gradient Boosting is selected on the basis of its marginally higher mean AUPR, its slightly lower variance across folds, and a nested estimate that closely tracks its test performance.

Decision-threshold tuning. For the selected Gradient Boosting model, the F1-optimal threshold estimated from out-of-fold training probabilities is 0.24, well below the default 0.5. Lowering the threshold trades precision for recall on the minority class, which is the desired behaviour for a campaign that aims to reach likely subscribers; the resulting test-set effect is analysed in Section 4.

4 Results

This section reports the final evaluation of the tuned models on the held-out test set (9043 instances, never used during selection or tuning). Following the model selection of Section 2.5, Gradient Boosting is the chosen model; we report all candidates for completeness and analyse the selected model in detail.

4.1 Final Model Comparison

Table 4 summarises test-set performance, ranked by AUPR. All models comfortably exceed the stratified baseline ($F1 = 0.171$) and the no-skill AUPR equal to the positive-class prevalence (≈ 0.175). Gradient Boosting attains the best AUPR (0.460) and ROC-AUC (0.789), confirming the nested-CV selection, with AdaBoost a close second. The kernel SVM and Logistic Regression are competitive on ranking metrics but no model breaks away: ROC-AUC lies within a narrow 0.777–0.789 band and AUPR within 0.435–0.460. This compression indicates a genuine performance ceiling once the leaky `duration` feature is removed — no remaining feature is individually strongly predictive — rather than a deficiency of any single model.

Model	Acc.	Bal. Acc.	F1	ROC-AUC	AUPR
Gradient Boosting	0.834	0.586	0.301	0.789	0.460
AdaBoost	0.835	0.581	0.287	0.786	0.458
Random Forest	0.788	0.690	0.471	0.777	0.443
Logistic Regression	0.717	0.705	0.460	0.779	0.438
SVM (RBF)	0.833	0.582	0.291	0.778	0.435
Grad. Boosting @0.24	0.787	0.700	0.482	0.789	0.460

Table 4: Final test-set performance, ranked by AUPR. The last row shows the selected model at its tuned decision threshold (0.24); threshold-independent metrics (ROC-AUC, AUPR) are unchanged by it.

The F1 and balanced-accuracy columns are not directly comparable across rows, because they depend on each model’s behaviour at the fixed 0.5 threshold. Models trained with `class_weight='balanced'` (Random Forest, Logistic Regression) operate aggressively and already reach high recall — and thus high F1 — at 0.5, whereas Gradient Boosting and AdaBoost, which lack a class-weight parameter, behave conservatively there. This is precisely why models are ranked by the threshold-independent AUPR rather than by F1 at an arbitrary cutoff: once Gradient Boosting is operated at its tuned threshold (last row), its F1 (0.482) surpasses every default-threshold model. Tuning also improves Random Forest over its default configuration (AUPR 0.443 here vs. 0.423 in Table 2), narrowing the gap to the boosting models.

A practical observation: Logistic Regression reaches a ROC-AUC (0.779) within one point of the best ensemble while being roughly an order of magnitude cheaper to train and fully interpretable through its coefficients. Where transparency or latency matter, it is a strong alternative to Gradient Boosting at negligible loss of ranking quality.

4.2 Effect of the Decision Threshold

At the default threshold of 0.5, Gradient Boosting maximises accuracy (0.834) but behaves conservatively on the minority class: its F1 is only 0.301 and its balanced accuracy 0.586, meaning many genuine subscribers are missed. Lowering the threshold to the F1-optimal value of 0.24 raises the positive-class F1 to 0.482 and balanced accuracy to 0.700, at the expected cost of overall accuracy (0.787). Because AUPR and ROC-AUC are threshold-independent, they are identical in both configurations; what the threshold changes is *where* on the precision–recall curve the model operates. Figure 1 shows the ROC and precision–recall curves for all models, and Figure 2 the confusion matrices of Gradient Boosting at both thresholds.

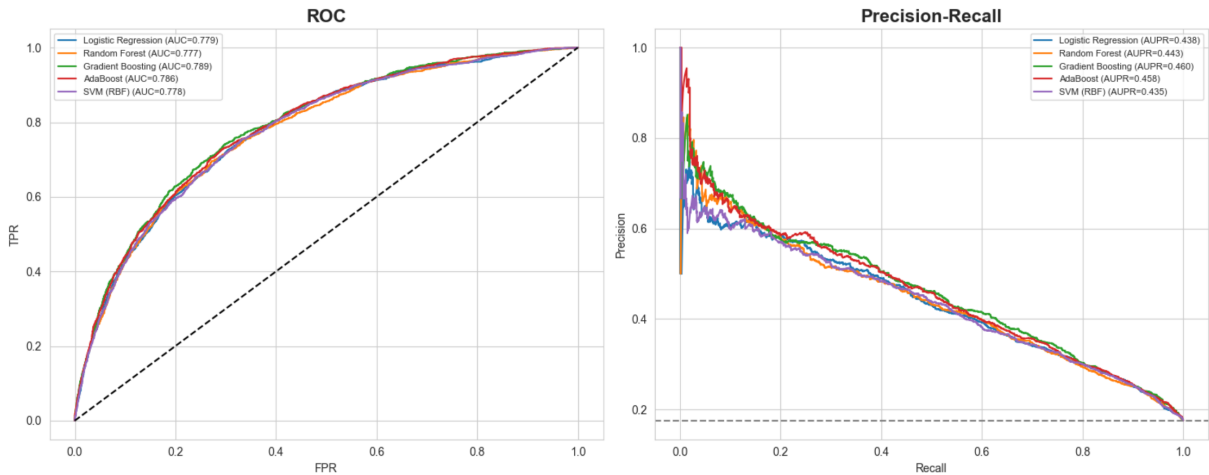


Figure 1: ROC (left) and precision–recall (right) curves on the test set. The dashed line in the PR plot marks the no-skill level (positive prevalence ≈ 0.175).

The per-class report of the selected operating point (Gradient Boosting at threshold 0.24) is given in Table 5. The model recovers 56.5% of true subscribers (positive recall) at a precision of 42.0%. For a telemarketing campaign whose objective is to identify clients worth calling, this recall-oriented trade-off is preferable to the high-precision, low-recall behaviour of the default threshold, since the cost of a missed subscriber (lost revenue) typically exceeds the cost of an additional call.

Class	Precision	Recall	F1	Support
no (0)	0.900	0.834	0.866	7457
yes (1)	0.420	0.565	0.482	1586
Macro avg	0.660	0.700	0.674	9043

Table 5: Per-class report of Gradient Boosting at the tuned threshold (0.24), test set.

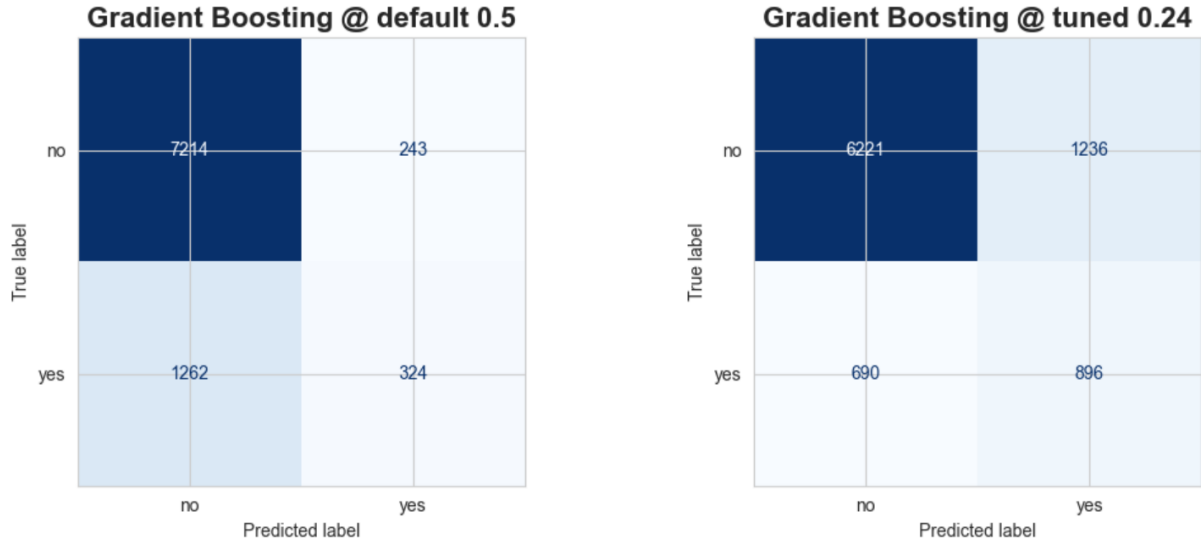


Figure 2: Confusion matrices of Gradient Boosting at the default threshold (0.5, left) and the tuned threshold (0.24, right). Lowering the threshold substantially increases the number of correctly identified subscribers.

4.3 The Cost of Avoiding Leakage

To quantify how much realistic performance is bounded by the exclusion of **duration**, the same logistic pipeline was evaluated with and without that feature. Including **duration** inflates performance to ROC-AUC 0.840 / AUPR 0.543, whereas the realistic configuration without it reaches the values reported for Logistic Regression in Table 4 (ROC-AUC 0.779 / AUPR 0.438). The ~ 0.06 ROC-AUC and ~ 0.11 AUPR gap is precisely the phantom performance that a leaking model would report. Our final results should therefore be read as a faithful estimate of *pre-call* predictability, which is the only setting in which such a model is deployable.

5 Conclusions

This project implemented a complete, leakage-free and reproducible ML pipeline on the Bank Marketing dataset, integrating unsupervised analysis, regularised linear models, kernel methods and ensembles. The emphasis was on methodological rigour and on understanding the limits of the signal rather than on maximising a single headline number.

Gradient Boosting was selected as the final model (test AUPR 0.460, about $2.6\times$ the no-skill level; ROC-AUC 0.789), but the most informative result is the *narrowness* of the performance band: all families lie within ROC-AUC 0.777–0.789 and AUPR 0.435–0.460, and the two best are not statistically distinguishable ($p = 0.37$). Once the leaky `duration` feature is removed (which alone accounts for ~ 0.06 ROC-AUC and ~ 0.11 AUPR of phantom skill), performance is bounded by the intrinsic difficulty and weak per-feature signal of the data, not by model capacity. Boosting marginally outperforms bagging, the kernel SVM offers no advantage at a much higher cost, and Logistic Regression remains a strong, interpretable alternative at near-identical ranking quality.

Two further findings shaped the work. First, accuracy is meaningless under this imbalance, so selection was driven by AUPR and ROC-AUC and the operating threshold was tuned on out-of-fold probabilities, trading precision for recall to recover more than half of all true subscribers — the appropriate orientation for prioritising campaign contacts. Second, clustering showed only weak structure (silhouette 0.282 at $k = 2$) unrelated to the target (ARI = 0.064), confirming that a discriminative model is necessary. Trustworthy estimates relied on re-fitting all preprocessing inside every fold, nested cross-validation (whose estimate, 0.466, closely matched the test AUPR of 0.460), and tuning strictly on training data.

5.1 Limitations and Future Work

The kernel SVM was tuned on an 8000-sample subset, slightly underestimating its full-data performance; the statistical tests rest on only five non-independent folds and are indicative rather than conclusive; and the data is a static 2008–2010 snapshot of one institution, limiting generalisation. Future work could replace F1-based thresholding with a *cost-sensitive* criterion, calibrate probabilities (isotonic/Platt) for expected-value ranking, compare class weighting against resampling (SMOTE), evaluate modern boosting implementations (XGBoost, LightGBM), and engineer richer previous-campaign and temporal features. None is expected to overcome the performance ceiling, but each would refine practical usefulness.

Closing remark. A disciplined pipeline — honest about leakage, rigorous in validation, explicit about imbalance — yields a modest but *credible* model, and on this dataset the more valuable task is understanding the limits of the signal rather than inflating a headline metric.

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